HK IMO Training 2023–2024
Problem Set 16
For 3 February 2024

Problem 61. Let m and n be positive integers satisfying $(m, n!) = 1$. Prove that $n! \mid m^{n!} - 1$.

Problem 62. Let x, y, z be positive real numbers satisfying $xy + yz + zx = xyz$. Find the minimum value of

$$x^x yz + xy^y z + xyz^z.$$

Problem 63. In $\triangle ABC$, D and E are the feet of altitudes from B and C respectively, H is the orthocentre, and M is the midpoint of BC. The circumcircles of $\triangle DEM$ and $\triangle BCH$ intersect at P and Q, with P being on the same side of the line AH as B. Prove that the lines DE, PH, MQ are concurrent at a point on the circumcircle of $\triangle ABC$.

Problem 64. There are 10000 coins in 100 colours, with exactly 100 coins of each colour. The coins are distributed to 100 people randomly such that each person receives exactly 100 coins. In each move, two people may exchange one of their coins. Suppose it is possible to exchange the coins finitely many times such that each person has 100 coins of different colours in the end. Prove that this can be done in a way that no coin is exchanged more than once.